

YEAR 7 MATHEMATICS

Chapter 1

Integers

WORKSHEETS

1

How to use this booklet

1 There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

2 Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

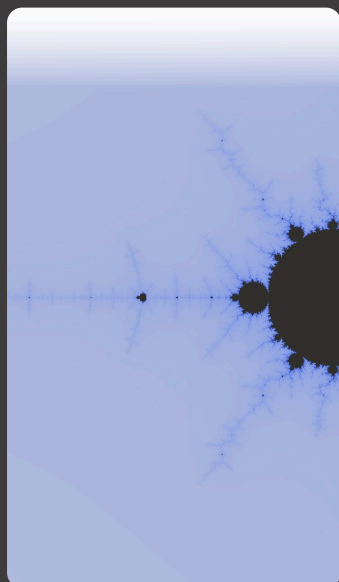
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

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In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers

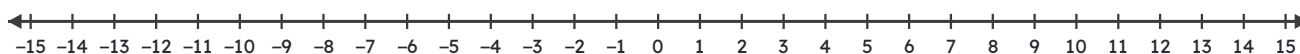


About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.



Easy

Questions 1–22

Write the opposite of each integer. It is the same distance from 0, on the other side.



WE DO With your teacher: copy the working.

E1 16

E2 7

YOU DO Your turn.

1 -59

59

2 41

3 9

4 47

5 -25

6 54

7 -14

8 -52

9 -43

10 23

11 27

Write an integer for each situation.



WE DO With your teacher: copy the working.

E3 63 floors up

YOU DO Your turn.

12 a fall of 8 points

-8

13 a fall of 48 points

14 50 m below sea level

15 52 minutes late

16 a gain of 4 kg

17 22 m below sea level

18 a loss of \$9

19 48 steps back

20 66 m above sea level

21 53 m below sea level

22 a loss of \$76

Medium

Questions 23–30

Simplify. The opposite of the opposite is the number itself.

WE DO With your teacher: copy the working.

E1 $-(-26)$

E2 $-(+10)$

YOU DO Your turn.

23 $-(-16)$

= the opposite of -16

= 16

24 $-(+36)$

= the opposite of 36

= ___

25 $-(-49)$

= the opposite of ___

= ___

26 $-(+7)$

27 $-(-57)$

28 $-(+2)$

29 $-(-25)$

30 $-(+26)$

Challenging

Questions 31–35

Answer each question. Write your reasoning on the lines.

WE DO With your teacher: copy the working.

E1 The opposite of a number is -16. What is the number?

E2 A diver is 27 m below sea level. Write this as an integer, then write its opposite.

E3 List all the integers whose distance from 0 is less than 4.

E4 If $n = -32$, find $-n$ and $-(-n)$.

YOU DO Your turn.

31 The opposite of a number is -34. What is the number?

opposite of ? = -34

the number is ___

32 A diver is 16 m below sea level. Write this as an integer, then write its opposite.

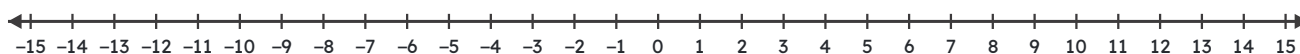
below sea level: ___

opposite: ___

33 List all the integers whose distance from 0 is less than 2.

34 If $n = -22$, find $-n$ and $-(-n)$.

35 The opposite of a number is -3. What is the number?



Easy

Questions 1–18

Which integer is further to the right on the number line? Write it on the line.

WE DO With your teacher: copy the working.

E1 5 or -4

E2 -2 or -3

YOU DO Your turn.

1 2 or -6

2

2 8 or -10

3 7 or -13

4 1 or -2

5 14 or -4

6 -2 or -11

7 -1 or 11

8 14 or 8

9 -13 or -7

10 3 or -12

11 10 or 1

Start at the first number and move along the number line. Where do you land?

WE DO With your teacher: copy the working.

E3 start at -1, move 7 right

YOU DO Your turn.

12 start at -5, move 4 left

-9

13 start at 3, move 7 right

14 start at 5, move 2 left

15 start at 5, move 4 left

16 start at 7, move 5 left

17 start at -7, move 4 right

18 start at 3, move 2 left

Medium

Questions 19–25

How many units apart are the two integers on the number line?

WE DO With your teacher: copy the working.

E1 -7 and 11

E2 -8 and 3

YOU DO Your turn.

19 3 and -1

-1 to 0 is 1, 0 to 3 is 3

1 + 3 = 4

20 11 and -12

-12 to 0 is 12, 0 to 11 is 11

___ + ___ = ___

21 12 and -13

___ to ___ is ___, ___ to ___ is ___

___ + ___ = ___

Which integer is exactly halfway between?

WE DO With your teacher: copy the working.

E3 -10 and -6

YOU DO Your turn.

22 -1 and 7

8 apart, half is 4

-1 + 4 = 3

23 -11 and -3

8 apart, half is 4

___ + ___ = ___

24 -16 and -2

___ apart, half is ___

___ + ___ = ___

25 -14 and 2

Challenging

Questions 26–30

Count the integers or find the halfway point. Show your working.

WE DO With your teacher: copy the working.

E1 How many integers are between -29 and 8?

E2 Which integer is halfway between -24 and 0?

YOU DO Your turn.

26 How many integers are between -4 and 23?

-3 to -1: 3 integers, and 0

1 to 22: 22 integers

___ + ___ + ___ = ___

27 Which integer is halfway between -15 and 15?

distance apart: ___

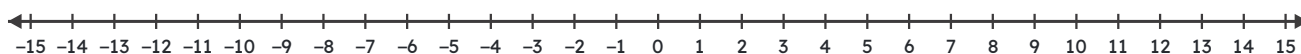
half: ___

___ + ___ = ___

28 How many integers are between -24 and 7?

29 Which integer is halfway between -13 and 13?

30 How many integers are between -19 and 30?



Easy

Questions 1–23

Write $<$ or $>$ in the box. The integer further right is larger.

WE DO With your teacher: copy the working.

E1 $-7 \square -8$

E2 $-5 \square -10$

YOU DO Your turn.

1 $-1 \square -20$

2 $-12 \square 12$

3 $7 \square -11$

4 $2 \square 10$

5 $11 \square 0$

6 $-9 \square -6$

7 $-4 \square -6$

8 $-1 \square -17$

9 $-5 \square -15$

10 $-10 \square -6$

11 $-10 \square 0$

Write the smaller integer.

WE DO With your teacher: copy the working.

E3 -7 or -2

YOU DO Your turn.

12 2 or -14

13 0 or -14

14 7 or -12

15 -21 or -1

16 6 or -26

17 -24 or -21

18 -14 or 3

19 -30 or -28

20 -30 or -17

21 10 or 9

22 -28 or 9

23 -30 or -19

Medium

Questions 24–30

Write the integers in ascending order (smallest first).

WE DO With your teacher: copy the working.

E1 $15, -8, -22, -9$

E2 $5, 14, -24, -18$

YOU DO Your turn.

24 $5, -20, 3, 0$

smallest: -20 , largest: 5

$-20, 0, 3, 5$

25 $-24, 5, 15, -6$

smallest: -24 , largest: 15

$____, ____ , ____ , ____$

26 $10, -7, -8, 4$

smallest: $____$, largest: $____$

$____, ____ , ____ , ____$

Write the integers in descending order (largest first).

WE DO With your teacher: copy the working.

E3 $12, -27, 25, -32$

YOU DO Your turn.

27 $9, -22, -16, -13$

largest: 9 , smallest: -22

$9, -13, -16, -22$

28 $-39, -23, -31, -2$

largest: -2 , smallest: -39

$____, ____ , ____ , ____$

29 $-6, 5, 9, 26$

largest: $____$, smallest: $____$

$____, ____ , ____ , ____$

30 $20, -31, -24, -23$

Challenging

Questions 31–36

Simplify any $-(-n)$ first, then answer the question.

WE DO With your teacher: copy the working.

E1 Write three integers greater than -14 but less than -9 .

E2 Which is the smallest: $-(-7)$, -7 or $-(-12)$?

E3 Write in ascending order: $-(-2)$, -9 , -12 , 0

YOU DO Your turn.

31 Write three integers greater than -18 but less than -12 .

integers from -17 to -13

e.g. $____, ____ , ____$

32 Which is the smallest: $-(-7)$, -8 or $-(-9)$?

$-(-7) = ____$, $-(-9) = ____$

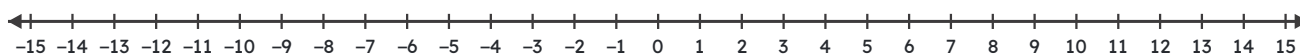
smallest: $____$

33 Write in ascending order: $-(-4)$, -7 , -3 , 0

34 Write three integers greater than -19 but less than -11 .

35 Which is the smallest: $-(-2)$, -6 or $-(-11)$?

36 Write in ascending order: $-(-9)$, -7 , -11 , 0



Easy

Questions 1–23

Evaluate each sum. Start at the first number and move along the number line.

WE DO With your teacher: copy the working.

E1 $-4 + 8$ _____

E2 $-7 + 10$ _____

YOU DO Your turn.

1 $-7 + 1$ -6

2 $-4 + 10$ _____

3 $-4 + 3$ _____

4 $-3 + 1$ _____

5 $-11 + 2$ _____

6 $-2 + 1$ _____

7 $-8 + 10$ _____

8 $-8 + 2$ _____

9 $-10 + 11$ _____

10 $-9 + 5$ _____

11 $-7 + 7$ _____

Evaluate. Adding a negative is the same as subtracting.

WE DO With your teacher: copy the working.

E3 $8 + (-1)$ _____

YOU DO Your turn.

12 $11 + (-1)$ 10

13 $-6 + (-7)$ _____

14 $10 + (-11)$ _____

15 $-4 + (-12)$ _____

16 $11 + (-4)$ _____

17 $12 + (-2)$ _____

18 $12 + (-3)$ _____

19 $7 + (-1)$ _____

20 $2 + (-11)$ _____

21 $-10 + (-6)$ _____

22 $-10 + (-11)$ _____

23 $-10 + (-1)$ _____

Medium

Questions 24–32

Evaluate. Add from left to right.

WE DO With your teacher: copy the working.

E1 $-12 + (-11) + (-13)$ _____

E2 $-4 + (-3) + (-6)$ _____

YOU DO Your turn.

24 $-4 + 14 + (-5)$

$= 10 + (-5)$

$= 5$

25 $-7 + (-3) + 9$

$= -10 + 9$

$= \underline{\hspace{1cm}}$

26 $10 + 2 + 5$

$= \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}}$

27 $-15 + 4 + 1$

28 $3 + (-8) + (-5)$

29 $-13 + 7 + 4$

30 $13 + 9 + 2$

31 $6 + 4 + (-3)$

32 $4 + 6 + (-7)$

Challenging

Questions 33–36

Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 The temperature is -7°C and rises 8°C . What is the new temperature?

E2 A lift at level -3 goes up 6 floors. Which level is it on?

E3 Sam has \$35 and spends \$51 on credit. What is the balance?

E4 Find the missing number: $-12 + \square = 6$

YOU DO Your turn.

33 The temperature is -6°C and rises 16°C . What is the new temperature?

$-6 + 16$

$= \underline{\hspace{1cm}}^{\circ}\text{C}$

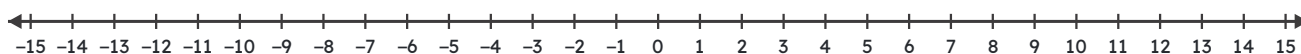
34 A lift at level -2 goes up 5 floors. Which level is it on?

$\underline{\hspace{1cm}} + \underline{\hspace{1cm}}$

$= \text{level } \underline{\hspace{1cm}}$

35 Sam has \$34 and spends \$62 on credit. What is the balance?

36 Find the missing number: $-5 + \square = 2$



Easy

Questions 1–23

Evaluate. Subtracting moves you left on the number line.

WE DO With your teacher: copy the working.

E1 $-6 - 4$ _____

E2 $-5 - 4$ _____

YOU DO Your turn.

1 $-8 - 7$ _____ **-15**

2 $-8 - 15$ _____

3 $2 - 2$ _____

4 $5 - 15$ _____

5 $5 - 13$ _____

6 $9 - 12$ _____

7 $6 - 10$ _____

8 $1 - 3$ _____

9 $-3 - 3$ _____

10 $0 - 8$ _____

11 $0 - 10$ _____

Evaluate. Subtracting a negative is the same as adding.

WE DO With your teacher: copy the working.

E3 $12 - (-9)$ _____

YOU DO Your turn.

12 $-2 - (-11)$ _____ **9**

13 $10 - (-1)$ _____

14 $8 - (-8)$ _____

15 $2 - (-9)$ _____

16 $-12 - (-5)$ _____

17 $3 - (-3)$ _____

18 $11 - (-10)$ _____

19 $-4 - (-8)$ _____

20 $-6 - (-1)$ _____

21 $-12 - (-3)$ _____

22 $12 - (-12)$ _____

23 $10 - (-7)$ _____

Medium

Questions 24–29

Evaluate. Change each $-$ ($-$) to $+$, then add from left to right.

WE DO With your teacher: copy the working.

E1 $1 - (-15) + (-15)$ _____

E2 $7 - (-13) + (-6)$ _____

YOU DO Your turn.

24 $-13 - (-7) + (-7)$

$= -13 + 7 + (-7)$

$= -6 + (-7)$

$= -13$

25 $-5 - (-1) + 15$

$= -5 + 1 + 15$

$= -4 + 15$

$= \underline{\hspace{1cm}}$

26 $14 - (-10) + 11$

$= \underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}}$

27 $13 - (-7) + (-5)$ _____

28 $6 - (-9) + (-5)$ _____

29 $9 - (-7) + (-7)$ _____

Challenging

Questions 30–33

Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 The minimum temperature was -5°C and the maximum was 19°C . By how much did it rise?

E2 A submarine at -62 m rises 22 m. Where is it now?

E3 What is 2 less than -9 ?

E4 Find the difference between -6 m and 24 m.

YOU DO Your turn.

30 The minimum temperature was -10°C and the maximum was 11°C . By how much did it rise?

$11 - (-10)$

$= 11 + 10$

$= \underline{\hspace{1cm}}^{\circ}\text{C}$

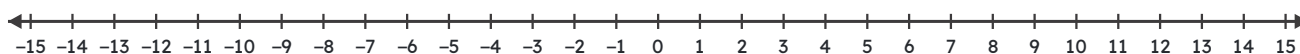
31 A submarine at -105 m rises 22 m. Where is it now?

$\underline{\hspace{1cm}} + \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}} \text{ m}$

32 What is 9 less than -4 ?

33 Find the difference between -15 m and 18 m.



Easy

Questions 1–23

Evaluate each product.
Same signs: positive.
Different signs: negative.

WE DO With your teacher: copy the working.

E1 -5×3

E2 $6 \times (-2)$

YOU DO Your turn.

1 $-7 \times (-9)$

63

2 -7×7

3 $-3 \times (-1)$

4 $1 \times (-3)$

5 $-1 \times (-7)$

6 $8 \times (-9)$

7 $4 \times (-9)$

8 -8×8

9 -10×8

10 $5 \times (-1)$

11 $-7 \times (-8)$

Write the sign of the answer only: + or -.

WE DO With your teacher: copy the working.

E3 $2 \times (-7)$

YOU DO Your turn.

12 $(-1) \times 9$

-

13 3×1

14 2×5

15 $(-2) \times 1$

16 $(-3) \times 3 \times 4$

17 7×7

18 $4 \times (-3)$

19 $1 \times (-1) \times 3$

20 $9 \times (-6) \times 6$

21 $7 \times 7 \times 3$

22 2×6

23 $9 \times (-9) \times (-8)$

Medium

Questions 24–41

Evaluate. Multiply two numbers at a time.

WE DO With your teacher: copy the working.

E1 $4 \times (-5) \times (-3)$

E2 $1 \times (-1) \times 4$

YOU DO Your turn.

24 $-6 \times (-2) \times (-3)$

$= 12 \times (-3)$

$= -36$

25 $5 \times (-1) \times (-2)$

$= -5 \times (-2)$

$= 10$

26 $-6 \times 5 \times (-5)$

$= ___ \times (___)$

$= ___$

27 $-1 \times (-1) \times (-1)$

28 $-3 \times 4 \times 5$

29 $3 \times (-4) \times 1$

30 $-3 \times 3 \times 2$

31 $-1 \times 1 \times 2$

32 $5 \times (-5) \times 4$

33 $-3 \times 6 \times (-1)$

34 $5 \times 4 \times (-5)$

35 $5 \times (-4) \times 1$

36 $-4 \times (-1) \times 4$

37 $-5 \times 1 \times (-1)$

38 $-4 \times 6 \times (-1)$

39 $-1 \times (-5) \times (-3)$

40 $2 \times (-4) \times 1$

41 $-6 \times 5 \times (-4)$

Challenging

Questions 42–46

Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 A diver goes down 2 m every minute. Where is she after 3 minutes?

E2 The temperature drops 4°C each hour for 3 hours. What is the total change?

E3 Find the missing number: $-4 \times \square = -12$

YOU DO Your turn.

42 A diver goes down 2 m every minute. Where is she after 6 minutes?

-2×6

$= ___ \text{ m}$

43 The temperature drops 6°C each hour for 6 hours. What is the total change?

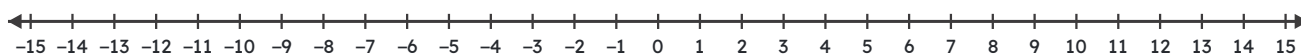
$___ \times ___$

$= ___ ^\circ\text{C}$

44 Find the missing number: $-4 \times \square = 32$

45 A diver goes down 4 m every minute. Where is she after 7 minutes?

46 The temperature drops 5°C each hour for 9 hours. What is the total change?



Easy

Questions 1–23

Evaluate each quotient.
Same signs: positive.
Different signs: negative.

WE DO With your teacher: copy the working.

E1 $36 \div (-4)$ _____

E2 $-28 \div (-7)$ _____

YOU DO Your turn.

1 $8 \div (-8)$ -1

2 $90 \div (-9)$ _____

3 $-4 \div (-1)$ _____

4 $40 \div (-8)$ _____

5 $4 \div (-1)$ _____

6 $-40 \div (-5)$ _____

7 $-56 \div 7$ _____

8 $40 \div (-5)$ _____

9 $-30 \div (-10)$ _____

10 $-12 \div 3$ _____

11 $-4 \div (-4)$ _____

Find the missing number.

WE DO With your teacher: copy the working.

E3 $30 \div \square = -5$ _____

YOU DO Your turn.

12 $36 \div \square = 4$ 9

13 $8 \div \square = 8$ _____

14 $25 \div \square = 5$ _____

15 $-14 \div \square = 7$ _____

16 $-5 \div \square = 5$ _____

17 $-5 \div \square = -5$ _____

18 $-9 \div \square = 9$ _____

19 $-20 \div \square = 4$ _____

20 $-21 \div \square = 7$ _____

21 $12 \div \square = -4$ _____

22 $-20 \div \square = -5$ _____

23 $49 \div \square = 7$ _____

Medium

Questions 24–41

Evaluate. Divide from left to right.

WE DO With your teacher: copy the working.

E1 $8 \div 4 \div 2$ _____

E2 $-40 \div (-5) \div (-2)$ _____

YOU DO Your turn.

24 $-10 \div 1 \div 2$

$= -10 \div 2$

$= -5$

25 $-9 \div 1 \div 3$

$= -9 \div 3$

$= -3$

26 $8 \div (-1) \div (-4)$

$= ___ \div (___)$

$= ___$

27 $-8 \div 2 \div 4$

28 $-8 \div 2 \div (-2)$

29 $-15 \div (-5) \div 3$

30 $-50 \div (-5) \div (-2)$

31 $-32 \div (-4) \div (-4)$

32 $45 \div 5 \div 3$

33 $10 \div 1 \div (-2)$

34 $4 \div 4 \div 1$

35 $32 \div (-4) \div 4$

36 $16 \div 4 \div 2$

37 $-60 \div 3 \div (-4)$

38 $36 \div 3 \div 4$

39 $6 \div (-2) \div 1$

40 $24 \div (-3) \div (-4)$

41 $-6 \div (-1) \div 3$

Challenging

Questions 42–46

Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 A debt of \$160 is shared equally by 8 people. Write each share as an integer.

E2 The temperature fell 15°C over 3 hours. What was the change each hour?

E3 Find the missing number: $\square \div 8 = 4$

YOU DO Your turn.

42 A debt of \$114 is shared equally by 6 people. Write each share as an integer.

$-114 \div 6$

$= ___ \text{ each}$

43 The temperature fell 12°C over 6 hours. What was the change each hour?

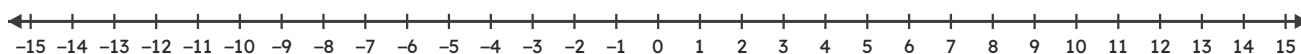
$___ \div ___$

$= ___^\circ\text{C each hour}$

44 Find the missing number: $\square \div (-4) = -5$

45 A debt of \$126 is shared equally by 7 people. Write each share as an integer.

46 The temperature fell 24°C over 4 hours. What was the change each hour?



Easy

Questions 1–23

Evaluate. $\times$ and $\div$ before $+$ and $-$.



WE DO With your teacher: copy the working.

E1 $3 + (-6) \times (-6)$ _____

E2 $-9 - 5 \times (-6)$ _____

YOU DO Your turn.

1 $-5 - (-6) \times (-4)$ -29

2 $-12 - (-2) \times (-2)$ _____

3 $-6 - (-5) \times (-6)$ _____

4 $10 + 6 \times 4$ _____

5 $2 + 4 \times 5$ _____

6 $-3 - (-4) \times 4$ _____

7 $7 + (-6) \times (-2)$ _____

8 $9 - 2 \times (-2)$ _____

9 $3 - (-4) \times 4$ _____

10 $9 + 4 \times 2$ _____

11 $8 + 5 \times (-6)$ _____

Evaluate. Brackets first.



WE DO With your teacher: copy the working.

E3 $(-6 - (-1)) \times 2$ _____

YOU DO Your turn.

12 $(2 + (-9)) \times 3$ -21

13 $(-5 - 7) \times 4$ _____

14 $(6 - 3) \times 3$ _____

15 $(2 + (-4)) \times 1$ _____

16 $(4 - (-9)) \times 1$ _____

17 $(7 - (-1)) \times (-4)$ _____

18 $(-5 - (-6)) \times 5$ _____

19 $(-6 + 4) \times (-2)$ _____

20 $(-8 - 1) \times 5$ _____

21 $(4 + (-9)) \times 1$ _____

22 $(-6 - 7) \times 4$ _____

23 $(-5 - 7) \times 1$ _____

Medium

Questions 24–32

Evaluate. Do $\times$ and $\div$ first (left to right), then $+$ and $-$.

WE DO With your teacher: copy the working.

E1 $6 \times 1 + 2 \div 1$ _____

E2 $3 \times (-6) + (-16) \div 4$ _____

YOU DO Your turn.

24 $-5 \times (-6) - (-5) \div (-5)$
= $30 - 1$
= 29

25 $-1 \times (-2) + (-10) \div 2$
= $2 + (-5)$
= -3

26 $1 \times 6 + (-10) \div 2$
= $6 + (-5)$
= 1

27 $1 \times 3 + 6 \div (-3)$ _____

28 $4 \times 1 + 15 \div 5$ _____

29 $6 \times (-4) - 8 \div 4$ _____

30 $6 \times (-2) - 20 \div 4$ _____

31 $-5 \times (-3) + 5 \div (-5)$ _____

32 $-6 \times 6 - 6 \div (-3)$ _____

Challenging

Questions 33–38

Evaluate. Brackets first, then $\times$ and $\div$, then $+$ and $-$. One step on each line.



WE DO With your teacher: copy the working.

E1 $7 - (-4 + 6) \times (-3)$ _____

E2 $6 - (-1 + (-2)) \times (-1)$ _____

YOU DO Your turn.

33 $-5 - (7 + 6) \times (-3)$
= $-5 - 13 \times (-3)$
= $-5 - (-39)$
= 34

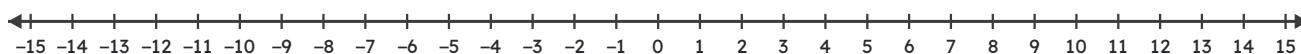
34 $5 - (2 + 8) \times (-1)$
= $5 - 10 \times (-1)$
= $5 - (-10)$
= 15

35 $-5 - (5 + 8) \times (-3)$ _____

36 $2 - (7 + (-4)) \times (-4)$ _____

37 $7 - (8 + (-1)) \times 3$ _____

38 $-7 - (-4 + 5) \times (-3)$ _____



Easy

Questions 1–23

Use your calculator. Use the $(-)$ key for negative numbers.



WE DO With your teacher: copy the working.

E1 $252 + (-391)$ _____

E2 $-369 + 350$ _____

YOU DO Your turn.

1 $27 + 17$ 44

2 $-160 + (-124)$ _____

3 $24 + 106$ _____

4 $-198 + 77$ _____

5 $-269 + 255$ _____

6 $269 + (-274)$ _____

7 $267 + 146$ _____

8 $-158 + 224$ _____

9 $-396 + 241$ _____

10 $197 + (-106)$ _____

11 $59 + (-266)$ _____

Use your calculator.



WE DO With your teacher: copy the working.

E3 $43 \times (-4)$ _____

YOU DO Your turn.

12 22×3 66

13 $45 \times (-44)$ _____

14 $3 \times (-11)$ _____

15 $-45 \times (-20)$ _____

16 -22×12 _____

17 $17 \times (-33)$ _____

18 $31 \times (-32)$ _____

19 $13 \times (-33)$ _____

20 $33 \times (-13)$ _____

21 -20×5 _____

22 $8 \times (-24)$ _____

23 45×43 _____

Medium

Questions 24–32

Use your calculator. Enter the brackets, then check the answer makes sense.

WE DO With your teacher: copy the working.

E1 $(-86 - (-81)) \times (-16)$ _____

E2 $(-3 - 67) \times (-11)$ _____

YOU DO Your turn.

24 $(-86 - (-8)) \times (-22)$ _____

bracket: -78

$-78 \times (-22) = 1\,716$

25 $(12 - 22) \times 19$ _____

bracket: -10

$___ \times ___ = ___$

26 $(12 - (-59)) \times (-18)$ _____

bracket: ___

$___ \times (___) = ___$

27 $(-44 - 62) \times (-21)$ _____

28 $(45 - 80) \times (-4)$ _____

29 $(-90 - 59) \times (-19)$ _____

30 $(-31 - 52) \times (-15)$ _____

31 $(57 - 83) \times 15$ _____

32 $(-74 - (-90)) \times (-3)$ _____

Challenging

Questions 33–39

Estimate first, then use your calculator. Is your answer close to the estimate?

WE DO With your teacher: copy the working.

E1 Estimate -60×44 , then calculate.

E2 Estimate -32×48 , then calculate.

YOU DO Your turn.

33 Estimate -34×17 , then calculate.

estimate: $-30 \times 20 \approx -600$

calculator:

34 Estimate -58×36 , then calculate.

estimate: $___ \times ___ \approx ___$

calculator:

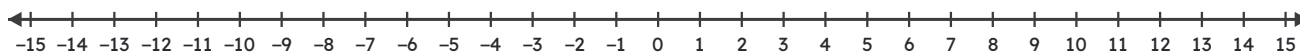
35 Estimate -42×35 , then calculate.

36 Estimate -46×28 , then calculate.

37 Estimate -38×30 , then calculate.

38 Estimate -57×13 , then calculate.

39 Estimate -48×26 , then calculate.



Easy

Questions 1–23

Find the new temperature.

WE DO With your teacher: copy the working.

E1 5°C, rises 14°C

E2 14°C, falls 4°C

YOU DO Your turn.

1 -4°C, falls 17°C

-21°C

2 -7°C, falls 16°C

3 -9°C, rises 15°C

4 11°C, falls 7°C

5 7°C, falls 5°C

6 -10°C, rises 10°C

7 5°C, falls 11°C

8 -2°C, falls 15°C

9 11°C, rises 10°C

10 12°C, falls 13°C

11 -2°C, falls 12°C

Find the new bank balance.

WE DO With your teacher: copy the working.

E3 -\$72, deposit \$92

YOU DO Your turn.

12 -\$9, withdraw \$103

-\$112

13 -\$23, deposit \$24

14 \$133, deposit \$70

15 \$118, deposit \$53

16 -\$73, withdraw \$18

17 \$103, withdraw \$78

18 -\$9, deposit \$86

19 \$13, deposit \$42

20 -\$21, withdraw \$64

21 -\$12, withdraw \$112

22 -\$50, deposit \$28

23 \$93, withdraw \$37

Medium

Questions 24–30

Find the difference between the two heights. Write a number sentence.

WE DO With your teacher: copy the working.

E1 a hill 267 m above sea level and a reef 57 m below

E2 a hill 280 m above sea level and a reef 74 m below

YOU DO Your turn.

24 a hill 106 m above sea level and a reef 10 m below

106 - (-10)

= 116 m

25 a hill 66 m above sea level and a reef 19 m below

66 - (-19)

= ___ m

26 a hill 52 m above sea level and a reef 32 m below

___ - (___)

= ___ m

27 a hill 174 m above sea level and a reef 48 m below

28 a hill 388 m above sea level and a reef 57 m below

29 a hill 129 m above sea level and a reef 45 m below

30 a hill 28 m above sea level and a reef 88 m below

Challenging

Questions 31–33

Write a number sentence for each step, then answer the question.

WE DO With your teacher: copy the working.

E1 A diver is at -45 m. She rises 4 m a minute for 4 minutes. Where is she now?

E2 A game scores +4 for a win and -3 for a loss. Mia wins 3 and loses 5. What is her score?

E3 It is -2°C at 6 a.m. It warms 3°C an hour for 3 hours. What is the temperature?

YOU DO Your turn.

31 A diver is at -30 m. She rises 5 m a minute for 5 minutes. Where is she now?

5 × 5 = 25

-30 + 25

= ___ m

32 A game scores +4 for a win and -1 for a loss. Mia wins 4 and loses 4. What is her score?

___ × ___ = ___

___ × (___) = ___

___ + (___) = ___

33 It is -3°C at 6 a.m. It warms 5°C an hour for 6 hours. What is the temperature?